

Auteur: Raj Shah
Studierichting: Informatica
Oefeningenreeks 6A nr. 9.7

4 termen, som is 2 en som der kwadraten is 46.

$$\rightarrow \begin{cases} (x-v) + x + (x+v) + (x+2v) = 2 \\ (x-v)^2 + x^2 + (x+v)^2 + (x+2v)^2 = 46 \end{cases}$$

$$\Leftrightarrow \begin{cases} 4x + 2v = 2 \\ 4x^2 + 4xv + 6v^2 = 46 \end{cases}$$

$$\Leftrightarrow \begin{cases} v = 1 - 2x \\ 4x^2 + 4x(1 - 2x) + 6(1 - 2x)^2 = 46 \end{cases}$$

$$\Leftrightarrow \begin{cases} v = 1 - 2x \\ 4x^2 + 4x - 8x^2 + 6 - 24x + 24x^2 = 46 \end{cases}$$

$$\Leftrightarrow \begin{cases} v = 1 - 2x \\ 20x^2 - 20x + 6 = 46 \end{cases}$$

$$\Leftrightarrow \begin{cases} v = 1 - 2x \\ 20(x^2 - x - 2) = 0 \end{cases}$$

$$x_1 = -1 \text{ en } x_2 = 2$$

$$\text{als } x = -1 \Rightarrow v = 1 - 2(-1) \Leftrightarrow v = 3$$

$$\text{als } x = 2 \Rightarrow v = 1 - 2(2) \Leftrightarrow v = -3$$

$$RR_1 : (x = -1, v = 3)(-4, -1, 2, 5)$$

$$RR_2 : (x = 2, v = -3)(5, 2, -1, -4)$$

References